

Weisheng Dong

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Education

2004.9~2010.8: Ph. D in Circuit and System, Xidian University, Xi'an, China.
Thesis: *The Context-based Adaptive Image Modeling for Image Restoration*
2000.9~2004.6: B.S in Communication Engineering, Hua Zhong University of Science and Technology, Wu Han, China

Research Interest

Sparse and low-rank image representation, image denoising, super-resolution, compressive sensing, deep learning

Profession experiences

2016.7~present: Professor, School of Electronic Engineering, Xidian University, Xi'an, China.
2012.7~2016.6: Associate professor, School of Electronic Engineering, Xidian University, Xi'an, China.
2010.9~2012.6: Lecturer, School of Electronic Engineering, Xidian University, Xi'an, China.
2012.8~2013.2: Visiting researcher, Visual Computing group, Microsoft Research Asia (MSRA), host: Yi Ma
2009.1~2010.6: Research Assistant/Associate (2009.1-2010.6), Dept. of Computing, The Hong Kong Polytechnic University, Hong Kong

Awards

Top 10% Paper Awards: IEEE International Conf. on Image Processing, 2014
Excellent Ph. D. Thesis: Excellent Ph. D. thesis award of Xidian University, 2011
Best Paper Award: SPIE Conf. on Visual Comm. and Image Process., 2010 (first author)

Grants

Structured sparse representation for image restoration, National natural Sciences Foundation of China (NSFC), 250, 000 RMB, 2012.1~2014.12, PI.
Dictionary learning for depth images recovery with applications to 3D reconstruction, National natural Sciences Foundation of China (NSFC), 830, 000 RMB, 2015. 1~2018.12.
Sparse representation and image restoration, National natural Sciences Foundation of China

(NSFC), 1, 300, 000 RMB, 2017. 1~2019. 12.

Publications

Book chapter:

- [1] Weisheng Dong and Xin Li, “Sparsity-regularized image restoration: locality and convexity revisited,” in *Image Restoration: Fundamentals and Advances*, CRC Press, Bahadir Gunturk and Xin Li (Editors), 2012.

Journal papers:

- [1] **Weisheng Dong**, Guangming Shi, Xin Li, Jinjian Wu, and Zhenhua Guo, “Color-guided depth recovery via local structural and nonlocal low-rank regularization,” **IEEE Transactions on Multimedia**, in press, 2016.
- [2] **Weisheng Dong**, Fazuo Fu, Guangming Shi, and Xun Cao, Jinjian Wu, Guangyu Li, and Xin Li, “Hyperspectral Image Super-Resolution via Non-Negative Structured Sparse Representation”, **IEEE Transactions on Image Processing (TIP)**, vol. 25, no. 5, pp. 2337-2352, May 2016.
- [3] **Weisheng Dong**, G. Shi, Y. Ma, and X. Li, “Image Restoration via Simultaneous Sparse Coding: Where Structured Sparsity Meets Gaussian Scale Mixture,” **International Journal of Computer Vision (IJCV)**, vol. 114, no. 2, pp. 217-232, Sep. 2015.
- [4] **Weisheng Dong**, Xiaolin Wu, and Guangming Shi, "Sparsity fine tuning in Wavelet domain with application to compressive image reconstruction", **IEEE Trans. on Image Processing (TIP)**, vol. 23, no. 12, pp. 5249-5262, Dec. 2014.
- [5] **Weisheng Dong**, Guangming Shi, Xiaocheng Hu, and Yi Ma, "Nonlocal sparse and low-rank regularization for optical flow estimation," **IEEE Trans. on Image Processing (TIP)**, vol. 23, no. 10, pp. 4527-4538, 2014.
- [6] **Weisheng Dong**, Guangming Shi, Xin Li, Yi Ma, and Feng Huang, "Compressive sensing via nonlocal low-rank regularization", **IEEE Trans. on Image Processing (TIP)**, vol. 23, no. 8, pp. 3618-3612, Aug. 2014. (54 citations)
- [7] **Weisheng Dong**, Lei Zhang, Guangming Shi, and Xin Li, “Nonlocally centralized sparse representation for image restoration,” **IEEE Trans. on Image Processing (TIP)**, vol. 22, no. 4, pp. 1620-1630, Apr. 2013. (264 citations)
- [8] **Weisheng Dong**, Lei Zhang, Rastislav Lukac, and Guangming Shi, “Sparse representation based image interpolation with nonlocal autoregressive modeling,” **IEEE Trans. on Image Processing (TIP)**, vol. 22, no. 4, pp. 1382-1394, Apr. 2013. (137 citations)
- [9] **Weisheng Dong**, Guangming Shi, and Xin Li, “Nonlocal image restoration with bilateral variance estimation: a low-rank approach,” **IEEE Trans. on Image Processing (TIP)**, vol. 22, no. 2, pp. 700-711, Feb. 2013. (134 citations)
- [10] **Weisheng Dong**, Lei Zhang, Guangming Shi, and Xiaolin Wu, “Image deblurring and super-resolution by adaptive sparse domain selection and adaptive regularization,” **IEEE Trans. On Image Processing**, vol. 20, no. 7, pp. 1838-1857, July 2011. (544 citations)
- [11] Xiaolin Wu, **Weisheng Dong**, Xiangjun Zhang, and Guangming Shi, “Model-assisted adaptive recovery of compressed sensing with imaging applications,” **IEEE Trans. on Image Processing (TIP)**, vol. 21, no. 2, Feb. 2012. (31 citations)
- [12] **Weisheng Dong**, Guangming Shi, Xiaolin Wu, Lei Zhang, “A learning-based method for compressive image recovery,” *Journal of Visual Communication and Image Representation*, vol. 24, no. 7, pp. 1055-1063, 2013.

- [13] **Weisheng Dong**, Xiafang Yang, and Guangming Shi, "Compressive sensing via reweighted TV and nonlocal sparsity regularisation", *Electronic Letters*, vol. 49, no. 3, pp. 184-186, 2013.
- [14] **Weisheng Dong**, Guangming Shi, Xin Li, Lei Zhang, and Xiaolin Wu, "Image reconstruction with locally adaptive sparsity and nonlocal robust regularization," *Signal Processing: Image Communication*, vol. 27, pp. 1109-1122, 2012. (33 citations)
- [15] **Weisheng Dong**, Guangming Shi, and Jizheng Xu, "Adaptive nonseparable interpolation for image compression with directional wavelet transform," *IEEE Signal Processing Letters*, vol. 15, pp. 233-236, 2008.
- [16] Ying Fu and **Weisheng Dong**, "3D Magnetic Resonance Image Denoising using Low-rank Tensor Approximation", *Neurocomputing*, Feb. 2016.
- [17] Lei Zhang, **Weisheng Dong**, Xiaolin Wu, and Guangming Shi "Spatial-temporal color video reproduction from noisy CFA sequence," *IEEE Trans. On Circuits and Systems for Video Technology*, vol. 20, no. 6, pp. 838-847, June 2010.
- [18] Lei Zhang, **Weisheng Dong**, David Zhang, Guangming Shi, "Two-stage Image Denoising by Principle Component Analysis with Local Pixel Grouping", *Pattern Recognition*, vol. 43, pp. 1531-1549, Apr. 2010. (325 citations)
- [19] Guangming Shi, **Weisheng Dong**, Xiaolin Wu, and Lei Zhang, "Context-based adaptive image resolution upconversion," *Journal of Electronic Imaging*, vol. 19, 013008, 2010.
- [20] **Weisheng Dong**, Guangming Shi, and Li Zhang, "Immune Memory clonal selection algorithms for designing stack filters," *Neurocomputing*, pp. 777-784, Jan. 2007.

Conference papers

- [21] Yongbo Li, **Weisheng Dong***, Xuemei Xie, Guangming Shi, Xin Li, and Donglai Xu, "Learned sparse models for image super-resolution," *Annual Conf. on Neural Information Processing Systems (NIPS)*, 2016.
- [22] Yongbo Li, **Weisheng Dong***, Guangming Shi, and Xuemei Xie, "Learning parametric distributions for image super-resolution: where patch matching meets sparse coding," in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2015.
- [23] **Weisheng Dong**, Guangyu Li, Guangming Shi, Xin Li, and Yi Ma, "Low-rank tensor approximation with Laplacian scale mixture modeling for multiframe image denoising", in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2015.
- [24] **Weisheng Dong**, Xin Li, Yi Ma, an Xin Li, "Image reconstruction via Bayesian Structured Sparse Coding", *IEEE Int. Conf. on Image Processing*, 2014. (Oral)
- [25] **Weisheng Dong**, Xiaolin Wu, and Guangming Shi, "Sparsity fine tuning in Wavelet domain with application to compressive image reconstruction", *IEEE Int. Conf. on Acoustics, Speech, and Signal Processing (ICASSP)*, 2014.
- [26] **Weisheng Dong**, Guangming Shi, and Xin Li, "Image deblurring with low-rank approximation structured sparse representation," *APSIPA*, 2012.
- [27] **Weisheng Dong**, Lei Zhang, and Guangming Shi, "Centralized sparse representation for image restoration", *IEEE International Conference on Computer Vision (ICCV)*, pp. 1259-1266, 2011. (123 citations)
- [28] **Weisheng Dong**, Xin Li, Lei Zhang, and Guangming Shi, "Sparsity-based image denoising via dictionary learning and structure clustering," *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 457-464, 2011. (Oral, acceptance rate: 3.5%, 219 citations)
- [29] **Weisheng Dong**, Guangming Shi, Lei Zhang, and Xiaolin Wu, "Super-resolution with nonlocal regularized sparse representation," in *Proc. SPIE Visual Communications and Image Processing (VCIP)*, 2010. (Best

Paper Award)

- [30] **Weisheng Dong**, Xin Li, Lei Zhang, and Guangming Shi, “Sparsity-based image deblurring with locally adaptive and nonlocally robust regularization,” accept to Proc. *IEEE International Conference on Image Processing* (ICIP), pp. 1841-1844, 2011.
- [31] **Weisheng Dong**, Xiaolin Wu, Guangming Shi, and Lei Zhang, “Context-based bias removal of statistical models of wavelet coefficients for image denoising,” in Proc. *IEEE International Conference on Image Processing* (ICIP), pp. 3841-3844, Oct. 2009.
- [32] **Weisheng Dong**, Lei Zhang, Guangming Shi, and Xiaolin Wu, “Nonlocal back-projection for adaptive image enlargement,” in Proc. *IEEE International Conference on Image Processing* (ICIP), pp. 349-352, Oct. 2009. (**113** citations)
- [33] Fangfang Wu, Guangming Shi, **Weisheng Dong**, and Xiaolin Wu, “Learning-based recovery of compressive sensing with application in multiple description coding,” in Proc. *IEEE International Workshop on Multimedia Signal Processing* (MMSP), Oct. 2009.
- [34] **Weisheng Dong**, Guangming Shi, and Jizheng Xu, “Signal-adapted directional lifting scheme for image compression,” in Proc. *IEEE International Symposium on Circuits and Systems* (ISCAS), pp. 1392-1395, 2008.

Professional Services

Associate Editor:

IEEE Transactions on Image Processing, 2015. 7 ~present

SIAM Journal on Imaging Sciences, 2017. 1~2019. 12

Circuit, System and Signal Processing, 2014. 9~present

Frontiers of Computer Sciences, 2015. 9~present

Technical Program Committee member of:

IEEE International Conference on Computer Vision and Pattern Recognition (CVPR) 2017, 2016, 2014、AAAI Conf. on Artificial Intelligence 2017, IEEE International Conference on Computer Vision (ICCV) 2015、European Conference on Computer (ECCV) 2016, 2014、SIGGRAPH Asia 2014、IEEE International Conference on Image Processing’ 2014, 2013, 2011, 2010、IEEE International Conference on Multimedia and Expo’ 2013, 2012、IEEE Conference on Visual Communication and Image Representation’ 2013, 2011.

Reviewer for:

IEEE TIP、IEEE TMI、IEEE TCSVT、IEEE TSMC-B、IEEE TMM、IJCV、IEEE SPL、SIAM J. on Imag. Sci.、Optics Express、Optics Letters、J. Bio. Opt.、Signal Processing、J. Electronic Imaging